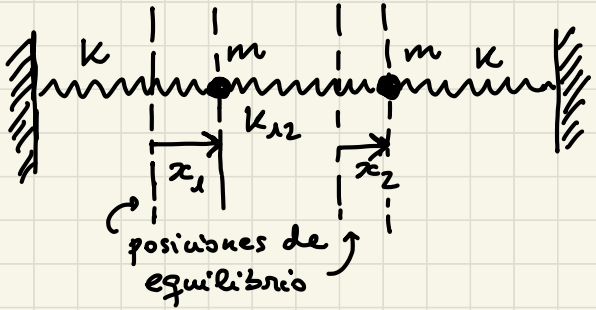
Ejemplo aplicación teoría general de osciladores



Coordenadas generalizadas:

$$q_1 = x_1, \quad q_2 = x_2$$

$$L = T - V = \frac{1}{2} m \dot{x}_1^2 + \frac{1}{2} m \dot{x}_2^2 - \frac{1}{2} k x_1^2 - \frac{1}{2} k x_2^2 - \frac{1}{2} k (x_2 - x_1)^2$$

$$L = \frac{1}{2} \sum_{j,k=1}^{2} m_{jk} \dot{q}_j \dot{q}_k - \frac{1}{2} \sum_{j,k=1}^{2} A_{jk} q_j q_k$$

Matriz $\{m\}$ $\quad T = \frac{1}{2} \sum_{j,k=1}^{2} m_{jk} \dot{q}_j \dot{q}_k = \frac{1}{2} m_{11} \dot{x}_1^2 + \frac{1}{2} m_{12} \dot{x}_1 \dot{x}_2 +$

$$+ \frac{1}{2} m_{21} \dot{x}_2 \dot{x}_1 + \frac{1}{2} m_{22} \dot{x}_2^2 =$$

$$= \frac{1}{2} m \dot{x}_1^2 + \frac{1}{2} m \dot{x}_2^2$$

$$\{m\} = \begin{pmatrix} m & 0 \\ 0 & m \end{pmatrix}$$

Matriz $[A]$ $\quad V = \frac{1}{2}\sum_{j,k=1}^{2} A_{jk} q_j q_k$

$$A_{jk} = \frac{\partial^2 V}{\partial q_j \partial q_k} \qquad V = \frac{1}{2}kx_1^2 + \frac{1}{2}kx_2^2 + \frac{1}{2}k_{12}(x_2 - x_1)^2$$

$$\frac{\partial V}{\partial x_1} = \frac{1}{2}k\cancel{2}x_1 - \frac{1}{2}k_{12}\cancel{2}(x_2 - x_1) = kx_1 - k_{12}(x_2 - x_1)$$

$$\frac{\partial V}{\partial x_2} = \frac{1}{2}k\cancel{2}x_2 + \frac{1}{2}k_{12}\cancel{2}(x_2 - x_1) = kx_2 + k_{12}(x_2 - x_1)$$

$$A_{11} = \frac{\partial^2 V}{\partial x_1 \partial x_1} = k + k_{12} \qquad A_{12} = \frac{\partial^2 V}{\partial x_1 \partial x_2} = -k_{12}$$

$$A_{22} = \frac{\partial^2 V}{\partial x_2 \partial x_2} = k + k_{12} \qquad A_{21} = \frac{\partial^2 V}{\partial x_2 \partial x_1} = -k_{12}$$

$$[A] = \begin{pmatrix} k + k_{12} & -k_{12} \\ -k_{12} & k + k_{12} \end{pmatrix}$$

Frecuencias características

$$\det\left[\underline{\underline{\zeta}}\,\underline{\underline{A}}\,\underline{\underline{\zeta}} - \omega^2\,\underline{\underline{\zeta}}\,\underline{\underline{m}}\,\underline{\underline{\zeta}}\right] = 0 \Rightarrow \begin{vmatrix} K + K_{12} - \omega^2 m & -K_{12} \\ -K_{12} & K + K_{12} - \omega^2 m \end{vmatrix} = 0$$

$$(K + K_{12} - \omega^2 m)^2 - K_{12}^2 = 0 \quad \text{Ec. característica}$$

$$K + K_{12} - \omega^2 m = \pm K_{12} \Rightarrow$$

$$\begin{cases} K + \cancel{K_{12}} - \omega^2 m = \cancel{K_{12}} \Rightarrow \omega^2 m = K \Rightarrow \omega_1 = \omega_S = \sqrt{\dfrac{K}{m}} \\ K + K_{12} - \omega^2 m = -K_{12} \Rightarrow \omega^2 m = K + 2K_{12} \Rightarrow \omega_2 = \omega_A = \sqrt{\dfrac{K + 2K_{12}}{m}} \end{cases}$$

Cualquier solución $x_1(t)$, $x_2(t)$ es combinación de:

$$\eta_1(t) = A_S \cos(\omega_S t - \delta_S) \quad . \quad \eta_2(t) = A_A \cos(\omega_A t - \delta_A)$$

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix} \qquad \vec{a}_1 = \begin{pmatrix} a_{11} \\ a_{21} \end{pmatrix} \quad \vec{a}_2 = \begin{pmatrix} a_{12} \\ a_{22} \end{pmatrix}$$

$\vec{a}_1$ y $\vec{a}_2$ verifican que: $[\eta A \eta - \omega_r^2 \eta m \eta]\vec{a}_r = 0$ ←

Modos normales

Modo $r=1$ oscila con $\omega_1 = \sqrt{\frac{k}{m}}$

Quiero que oscile $\eta_1(t)$ y que $\eta_2(t) = 0 \ \forall t$

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} \eta_1 \\ 0 \end{pmatrix} \Rightarrow \begin{cases} x_1 = a_{11}\eta_1 \\ x_2 = a_{21}\eta_1 \end{cases} \text{ donde } \vec{a}_1 = \begin{pmatrix} a_{11} \\ a_{21} \end{pmatrix} \rightarrow$$

$$\rightarrow [\eta A \eta - \omega_1^2 \eta m \eta]\vec{a}_1 = 0 \Rightarrow \begin{pmatrix} k + k_{12} - \frac{k}{m}m & -k_{12} \\ -k_{12} & k + k_{12} - \frac{k}{m}m \end{pmatrix} \begin{pmatrix} a_{11} \\ a_{21} \end{pmatrix} = 0$$

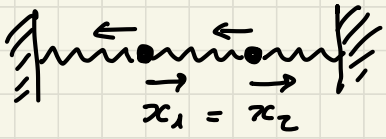
$$\begin{pmatrix} k_{12} & -k_{12} \\ -k_{12} & k_{12} \end{pmatrix}\begin{pmatrix} a_{11} \\ a_{21} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \begin{cases} k_{12}a_{11} - k_{12}a_{21} = 0 \\ -k_{12}a_{11} + k_{12}a_{21} = 0 \end{cases} \longrightarrow$$

$$\longrightarrow k_{12}a_{11} = k_{12}a_{21} \Rightarrow a_{11} = a_{21} = a \Rightarrow \vec{a}_1 = \begin{pmatrix} a \\ a \end{pmatrix} \Rightarrow$$

$$\left.\begin{aligned} x_1 &= a\eta_1 \\ x_2 &= a\eta_1 \end{aligned}\right\} \Rightarrow \boxed{x_1 = x_2}$$

modo normal 1

$$\omega_1 = \sqrt{\frac{k}{m}}$$



Modo normal $n=2$ oscila con $\omega_2 = \sqrt{\frac{k+2k_{12}}{m}}$

Quiero que $\eta_2(t)$ y $\eta_1(t) = 0$ $\forall t$

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} a_{11} & a_{21} \\ a_{21} & a_{22} \end{pmatrix}\begin{pmatrix} 0 \\ \eta_2 \end{pmatrix} \Rightarrow \left.\begin{aligned} x_1 &= a_{21}\eta_2 \\ x_2 &= a_{22}\eta_2 \end{aligned}\right\} \text{ donde } \vec{a}_2 = \begin{pmatrix} a_{21} \\ a_{22} \end{pmatrix} \longrightarrow$$

$$\longrightarrow [\eta A \eta - \omega_2^2 \eta m \eta]\vec{a}_2 = 0 \Rightarrow \begin{pmatrix} k + k_{12} - \frac{k+2k_{12}}{m}m & -k_{12} \\ -k_{12} & k + k_{12} - \frac{k+2k_{12}}{m}m \end{pmatrix}\begin{pmatrix} a_{21} \\ a_{22} \end{pmatrix} = 0$$

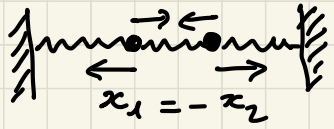
$$\Rightarrow \begin{pmatrix} -k_{12} & -k_{12} \\ -k_{12} & -k_{12} \end{pmatrix} \begin{pmatrix} a_{21} \\ a_{22} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \left.\begin{array}{l} -k_{12}a_{21} - k_{12}a_{22} = 0 \\ -k_{12}a_{21} - k_{12}a_{22} = 0 \end{array}\right\} \longrightarrow$$

$$\longrightarrow -k_{12}a_{21} = k_{12}a_{22} \Rightarrow a_{21} = -a_{22} = b \Rightarrow \vec{a}_2 = \begin{pmatrix} b \\ -b \end{pmatrix} \Rightarrow$$

$$\Rightarrow \left.\begin{array}{l} x_1 = b\eta_2 \\ x_2 = -b\eta_2 \end{array}\right\} \Rightarrow \boxed{x_1 = -x_2}$$

modo normal 2

$$\omega_2 = \sqrt{\frac{k+2k_{12}}{m}}$$



Relación entre $\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ y $\begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix}$

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} a & b \\ a & -b \end{pmatrix} \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix} \Rightarrow \left.\begin{array}{l} x_1 = a\eta_1 + b\eta_2 \\ x_2 = a\eta_1 - b\eta_2 \end{array}\right.$$

despejando η_1 y η_2 $\Rightarrow$

$$\Rightarrow \left.\begin{array}{l} \eta_1 = \dfrac{x_1 + x_2}{a} \\ \eta_2 = \dfrac{x_1 - x_2}{b} \end{array}\right.$$

(ver Marion pág. 487)

donde a y b son arbitrarios. Podemos tomar por ejemplo $a = b = 2$ y obtendríamos las mismas relaciones que en la primera parte de este tema